

Analysis of tourism information sources using a Mokken Scale perspective



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HIGHLIGHTS

- Mokken Scale Analysis (MSA) was used to evaluate scalability of information sources.
- A sample of 594 questionnaires is used to evaluate information search effort.
- Scalability of information sources was demonstrated, identifying two different scales.
- Results indicate some variations into scales considering socio-demographic and travel behavioural variables.

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ABSTRACT

This study contributes to the analysis of information sources as a determinant of the travel decision-making process. It focuses on the use of Mokken Scale Analysis (MSA) as a new methodological approach to detect unidimensional hierarchical scale in the analysis of information sources. The paper shows how MSA can be applied in the tourism field. Findings reveal two different scales (active and passive information search scales) for classifying information sources in tourism. Variations in the scales are revealed, taking socio-demographic and travel behavioural variables into consideration. The conclusions clearly explain the implications for tourism marketers and make recommendations for future research.

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1. Introduction

Understanding information search behaviour has been one of the most widely studied issues in the tourism field in recent decades (Fodness & Murray, 1999; Gitelson & Crompton, 1983; Llodrà-Riera, Martínez-Ruiz, Jiménez-Zarco, & Izquierdo-Yusta, 2015; Money & Crotts, 2003) especially as information sources have been postulated as one of the most influential factors in the decision-making process (Dey & Sarma, 2010).

Gursoy (2003) highlights the long tradition of tourism marketing literature by examining tourism information search behaviour, both conceptually and empirically. Former studies on information search behaviour have focused on identifying the most frequently used information sources (Fodness & Murray, 1999; Gitelson &

Crompton, 1983; Llodrà-Riera et al., 2015) by tourists at different stages in the decision-making process. These studies frequently try to determine the influence of an individual's socio-demographic characteristics and/or an individual's travel-specific differences (Bieger & Laesser, 2004). Fodness and Murray (1997) highlight the relevance of using a multiple source approach. Previous to their study, research had been centred on analysing users of single information sources. From this point of view, numerous studies have focused on analysing how many different types of sources were used (i.e. Gitelson & Crompton, 1983), and the interdependence among the various information sources tourists consult when organising their holidays (Gronflaten, 2009; Gursoy & Chen, 2000).

Until now researchers have outlined which types of information sources are most frequently used considering the various influencing factors mentioned above. However, there are no studies which detect whether specific information sources are a single latent dimension in a hierarchical scale.

Studying such a scale is important for tourism destination

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marketers to be able to determine the dimensions and hierarchy in which information sources are used. Studying the scale for socio-demographic and travel behaviour characteristics among users can be useful for understanding how different tourist profiles use the information sources and to identify whether differences in the scale structure exists between these characteristics. This information will give them a better understanding of the communication actions which should be carried out, so that they can be designed and executed more efficiently.

Therefore, the aim of this paper is twofold. First, to determine whether information sources in tourism belong to a single scale, and to detect if the items, which are represented by information sources are scalable, or hierarchically ordered, within the scale, Mokken Scale Analysis (MSA) will be used. Second, to conclude whether the resulting scale is comparable across socio-demographic and travel behavioural variables, in order to decide if communication strategy should be differentiated according to different market segments.

MSA technique has been extensively used in psychology for assessing the dimensionality and requirements for constructing scales (Meijer & Baneke, 2004; Smits, Timmerman, & Meijer, 2012), especially for personality traits (Wismeijer, Sijtsma, van Assen, & Vingerhoets, 2008), education (Vargas–Vargas, Jimenez, Meseguer-Santamaria, Montro-Lorenzo, & Fernandez-Aviles, 2010), political science (Van Schuur, 2003), marketing research (Paas & Molenaar, 2005), economics (Paas, 1998) and medical research (Roorda et al., 2005). In the tourism field, only Mazanec, Crotts, Gursoy, and Lu (2015) have made the first approach to this technique through their study of cultural values dimensions. Hence, the use of MSA in this article can be considered a contribution to the better understanding of information search behaviour. At the same time, it shows the adequacy of MSA in the tourism field in general, and particularly in information search behaviour.

This paper is organised as follows: the following section reviews the literature on tourism information search and the determinants and methods used for its analysis. In the next section sampling, data collection and data analysis are explained. Considering that Mokken Scale Analysis (MSA) is the main statistical method employed, the requirements needed to apply the method are described. This is followed by the results, and a final section where conclusions and adequacy of MSA for tourism research are discussed.

2. Theoretical background

2.1. Tourism information search

As in many products and services, consumer behaviour in tourism begins with acquiring and selecting information, in order to make the right decision according to individual needs. In this context, Sirakaya and Woodside (2005, p. 827) state “a tourist is expected to be highly involved in the information search for tourism service purchases, than many other product or service purchases, because of high-perceived risk factor”. Information search is the motivated activation of knowledge stored in our memory or the acquisition of information from the environment (Engel, Blackwell, & Miniard, 1995). This definition suggests that information sources can be internal or external. First, “internal information may have been acquired at one time from previous experience and past information searches, or passively through low-involvement learning where consumers are repeatedly exposed to marketing stimuli” (Money & Crotts, 2003, p. 193).

Second, external information is acquired from the environment, and this represents a conscious effort to seek out new information (Gitelson & Crompton, 1983). Along these lines, Gitelson and

Crompton (1983) highlight three main reasons to expect external searches in tourism. These are: (1) vacations are considered a high-risk purchase, (2) intangibility of the tourism product and (3) unfamiliarity with the new destination. According to Fodness and Murray (1997), external information sources can be classified in two ways: depending on whether the source is commercial or non-commercial; and whether it is received from a personal or impersonal communication, for a continuum of 11 different sources.

The types of information sources included vary substantially from one study to another. For example, Money and Crotts (2003, p. 193) detect four different categories: a) personal (e.g. word-of-mouth, or advice from friends and relatives), b) marketer-dominated (e.g. advertisements in print and electronic media), c) neutral (e.g. third-party sources such as travel agents and travel guides), and d) experiential sources—direct contacts with retailer; while Bieger and Laesser (2004) included 18 different types of information sources. In this context, Gitelson and Crompton (1983) used only 5 categories, based on the aggregation made by Nolan (1976). Furthermore, the emergence of Internet has increased the number of external information sources, and also the number of studies that differentiate between online and offline information searches (Li, Pan, Zhang, & Smith, 2009; Llodrà-Riera et al., 2015; Luo, Feng, & Cai, 2004).

Gronflaten (2009), on the other hand, highlights the need to differentiate between information sources (provider of the information) and channels (communication method). In this sense, Llodrà-Riera et al. (2015) advocate that Internet is a channel which comprises several information sources with different natures, and include 19 items in their study (e.g. search engines, official websites, forums, applications for sharing photos and videos, etc.). These are seen as separate from the 14 items classified as traditional information sources. In fact, there is no agreement on the number or type of variables that should be included when analysing information search behaviour.

In turn, information sources can also be classified according to whether the consumer is actively involved in the information search or not. Gunn (1972), in her seminal contribution, through the model of the seven-stages of travel experience, advocates the difference between organic and induced tourist images. These images are mainly differentiated by the information seeking behaviour of the tourist and whether this is passive or active. Gartner (1994), also points this out, asserting that word-of-mouth can be disseminated in an unsolicited manner.

Additionally, “numerous studies have emerged that attempt to assess the credibility of various sources of information ranging from advertising to word-of-mouth” (Mack, Blose, & Bing Pan, 2008, p. 135). Kerstetter and Cho (2004) argue that credibility of information sources is relevant to influencing decision-making process. In this regard, Gartner (1994) ranks credibility of various image formation agents, and highlights word-of-mouth from organic agents (solicited or unsolicited) as the most credible information source. And more recently, Camprubí, Guia, and Comas (2013) mention that the new Web 2.0 paradigm has maintained the credibility of organic agents, but it has increased its market penetration. Other recent studies confirm that tourists who share their travel experiences in virtual travel communities have high credibility among other members (Buhalis & Law, 2008). However Mack, Blose, and Pan (2008) concluded that blogs are less credible than face to face word-of-mouth communication.

In a nutshell, information search is a dynamic process (Bieger & Laesser, 2004), where individuals use various amounts and types of information sources, in response to internal and external information needs, to facilitate travel planning (D. Fodness & Murray, 1997). During this process, when there is insufficient internal information to make a decision, individuals move to seek external

information sources (D. Fodness & Murray, 1997; Gitelson & Crompton, 1983; Gursoy & Chen, 2000; Gursoy & McCleary, 2004). In this context, Bieger and Laesser (2004) worked on a first approach to determine different preferences for information sources depending on the stage at which decisions were made regarding the travel decision-making and travel preparation. In their seminal study they found that information sources used before or after making the decision to travel, were different, resulting in an information source path model. This is the first model showing that changes can be produced in the use of information sources from an evolutionary point of view. However, it is not likely to determine the order in which the information sources are used, as it forms part of the dynamic process of information seeking.

2.2. Determinants of information search behaviour

Several researchers have considered the relevance of determining the factors that influence information search behaviour. Wilson (1997) detected that several disciplines, including health communication, personality studies, consumer research and innovation studies have exhaustively discussed intervening variables in this issue. A set of variables and sub-variables were also identified. These were personal characteristics (emotional, educational and demographical variables), social/interpersonal variables and environmental variables (economic variables and source characteristics).

Regarding specific academic literature on tourism consumer behaviour, two types of variables are identified, socio-demographic and travel behavioural variables. First, there is a common agreement that socio-demographic characteristics are conditioning factors in information search strategy (Gronflaten, 2009). These characteristics include variables such as age, gender, income, education, nationality and occupation. For example, Luo et al. (2004, p. 22) confirmed that “gender and household income are related to information search preferences”. However, in the same study they also found that no differences exist among people of different ages, education levels or occupations. Fodness and Murray (1999) also detected that level of income affected information sources used. In particular, they found that tourists with a lower household income used recommendations from friends and relatives more; and those with higher incomes tended to use tourist offices to gather information. Focussing on gender and family status, Gitelson and Crompton (1983, p. 7) found that “family groups and females were more likely than other groups to receive information from radio or television”. Cross-cultural studies have also revealed that nationality influences some aspects of information search behaviour (Gursoy & Chen, 2000; Money & Crotts, 2003). For instance, Gursoy and Chen (2000) confirmed that the British and the French have similar information seeking behaviour, however substantial differences were found with Germans. Along the same lines, Money and Crotts (2003) determined differences between Germans and the Japanese.

Second, studies of information search behaviour also consider travel behavioural factors, including variables such as travel party composition, length of stay, purpose of the trip, number of previous visits, type of accommodation, distance travelled, planning horizon, and so on (Gronflaten, 2009). For instance, Fodness and Murray (1999) found that there is a relationship between length of stay and information search; in particular, Gitelson and Crompton (1983) determined that a high information search is associated with longer trips, and also the use of destination-specific literature, print media and travel consultants. In studies which focus on comparing online and offline information search, Luo et al. (2004, p. 22) concluded that “tourists who search on the Internet tend to

spend more at their destinations as compared to those who consult with other information sources”. The level of familiarity with the destination or the comparison between first-time visitors and repeaters is also examined in previous research (Bruwer & Thach, 2013; Gitelson & Crompton, 1983; Gursoy, 2003). Other studies have concluded that familiarity with the destination influences both types and degree of information sources used (Gursoy, 2003; Li et al., 2009), and Bruwer and Thach (2013) established that first-time visitors tend to use more sources in wine destinations, such as brochures and tourists offices, than repeaters.

In general terms, there is evidence to show that studies analysing information search behaviour combine both socio-demographic and travel behavioural factors. However, Bieger and Laesser (2004, p. 364) state that “trip-specific variables have a better explanatory value than socio-demographic variables”. In any case, such variables are complementary, considering the tendency of some studies to identify information search profiles of tourists (D. Fodness & Murray, 1997; Gitelson & Crompton, 1983).

2.3. Methods used to analyse information search

Analysis techniques employed in this field generally aim to determine the importance of information sources used. The relevance of information source effort is commonly defined through the frequency of use of the various information sources (Dale Fodness & Murray, 1999; Gitelson & Crompton, 1983), or the number of information sources used (Gitelson & Crompton, 1983), sometimes computed through the average of use intensity (Llodrà-Riera et al., 2015). Other studies aim to analyse the relationship between information sources and socio-demographic variables or travel behavioural variables. In this context, bivariate statistical techniques are the most frequently used, such as Chi-square test, t-test or analysis of variance (Bruwer & Thach, 2013; Luo et al., 2004).

A number of studies using multivariate techniques have also been identified. These techniques are normally used with two different purposes. On the one hand, they attempt to aggregate the information sources in relation to various similarity parameters, such as Principal Components Analysis (PCA). In this case, Llodrà-Riera et al. (2015) use this technique to differentiate between groups of traditional information sources and online information sources. On the other hand, a cluster analysis is used to determine the various users' profiles in the information search. For instance, Bieger and Laesser (2004) carried out cluster analysis with the purpose of finding different behavioural profiles in the information search, before and after deciding the holiday destination. Other studies use multivariate techniques as an indirect method, such as that by Gitelson and Crompton (1983). In particular, these authors use a Factor Analysis (FA) on the basis of tourist motivation, and later determine the relationship among the three resulting factors and the information sources used through a Duncan's multiple range test. In fact, the combination of more than one statistical technique to carry out a complete analysis is common, as seen in several studies (i.e., Gitelson & Crompton, 1983; Gronflaten, 2009).

The methods used in the literature have enabled the identification of the most relevant information sources, the average of use intensity, or the aggregation of used information sources by means of multivariate techniques, such as PCA or FA. However, techniques which determine scalability in groups of information sources based on tourist behaviour.

have not been used. Thus, this article is based on a new approximation to the analysis of information sources; the use of Mokken Scale (Mokken, 1971; Sijtsma & Molenaar, 2002; Van Schuur, 2011). This technique is applied to a set of information sources, in order to investigate the homogeneity and scalability of the set. This is useful for researchers in the sense that they will be

able to classify and sort different information sources according to their ‘popularity’ of use for tourists.

MSA represents new challenges in creating scales in tourism research, since PCA is widely used in this field (Calantone, Di Benedetto, Hakam, & Bojanic, 1989; Kim & Richardson, 2003; Walmsley & Young, 1998). The difference between MSA and PCA is that the later assumes linearity in the relationship between item scores and underlying traits. “PCA is suited to dimensionality analysis, but it is not a measurement model that implies particular scale properties” (Wismeijer et al., 2008, p. 329). In another way, “MSA not only allows for dimensionality analysis, but also investigates the fit of a measurement model to the data” (Wismeijer et al., 2008, p. 329). It is able to reject an item or a group of items on a measurement scale, which means finding items that are not scalable. Thus, “MSA helps moving measurement and theory building forward” (Wismeijer et al., 2008, p. 329). Several studies compared MSA method with PCA (Van Schuur, 2003; Waller, Tellegen, McDonald, & Lykken, 1996; Wismeijer et al., 2008) or FA (Emons, Sijtsma, & Pedersen, 2012), which is out of the scope of this article. Considering the characteristics of MSA, this technique can be properly used in the tourism field, and more specifically in information source analysis.

3. Methodology

3.1. Sampling and data collection

Data collection was carried out in Palafrugell, a municipality located on the Costa Brava, in the region of Catalonia (Spain). Five different towns make up this municipality, three of them located on the coastline. Therefore, this destination concentrates several tourism resources, such as natural landscapes, beaches and heritage sites. The tourism board is estimated to have received more than 61,000 visitors in 2013, especially concentrated during the summer months.

The period for data collection was between April and September 2014. The initial sample was 600 respondents (4% margin of error with 95% confidence level), after data cleaning, 6 cases were eliminated for validity reasons. Thus, the final sample size was 594 respondents and data were collected using a survey on tablet computers. The sample was stratified in the different months, according to the distribution of the number of tourists in each period. Once the number of daily interviews was determined, a systematic sampling technique was performed by selecting the fifth person after each interview.

The questionnaire contained the following topics: demand profile, travel arrangements, information sources, perceived image of the destination and satisfaction. The specific question used

concerning information sources (Fig. 1) was: “To what degree have you used the following information sources to find out about this destination? (1 means used very little; 5 means used a lot)”, the category “not used” was also available for the respondents. Information sources used were as follows: official website, tourism blogs, search engines (such as Google), travel agency, tourism fairs, tourism brochures, mass media (press, radio and television), and recommendations from friends and relatives.

The sample respondents’ profile (gender, age, education and place of residence) is shown in Table 1. Demographic information shows that there are slightly more females than males. The age range of most tourists is from 45 to 54 years old, followed by tourists between 35 and 44 years old, the majority having university studies. This is a destination with 37.6% national and 62.4% international tourists.

3.2. Mokken Scale Analysis (MSA)

There are different methods to analyses cumulative scales. A perfect deterministic cumulative scale (Guttman, 1950) would indicate that, in this case, if information sources are hierarchically ordered for all tourists from high popularity (generally more used) to low popularity, people who use information sources with lower popularity will also use all information sources with greater popularity.

However, such perfect deterministic cumulative scales are unlikely to exist, because all concepts do not necessarily satisfy these criteria. In order to relax these strict conditions, the so called Mokken scale (Mokken, 1971, 1997; Sijtsma & Molenaar, 2002; Van

Table 1
Profile of respondents.

	Frequency	%
<i>Gender</i>		
Male	243	41.3%
Female	346	58.7%
<i>Age</i>		
Less than 35	97	16.4%
35 to 44	151	25.5%
45 to 54	186	31.5%
55 to 64	101	17.1%
More than 64	56	9.5%
<i>Education</i>		
Primary	35	5.9%
Secondary	216	36.7%
University	338	57.4%
<i>Place of residence</i>		
Nationals	222	37.6%
Internationals	368	62.4%

To what degree have you used the following information sources to find out about this destination?
(1 means not much use; and 5 means used a lot)

	1	2	3	4	5	Not used
Official website	☐	☐	☐	☐	☐	☐
Tourism blogs	☐	☐	☐	☐	☐	☐
Search engines (Google)	☐	☐	☐	☐	☐	☐
Travel agency	☐	☐	☐	☐	☐	☐
Tourism fairs	☐	☐	☐	☐	☐	☐
Tourism brochures	☐	☐	☐	☐	☐	☐
Press, radio and television	☐	☐	☐	☐	☐	☐
Friends and relatives recommendation	☐	☐	☐	☐	☐	☐

Fig. 1. Question on information sources used.

Schuur, 2011) can be used. It is a non-parametric method based on Item Response Theory, where assumptions about the scale's dimensionality and characteristics of items hierarchy (cumulative scale or ranking of items in a scale) are tested (Embretson & Reise, 2000; Van der Linden & Hambleton, 1997).

MSA is a model that is used for scaling items and scoring respondents on an ordinal scale, which can also be used with a small number of items (Van Schuur, 2003, p. 139). The idea behind MSA is that the probability of a positive response to an item can be seen as a function of the subject's latent trait score (θ) and the properties of the item (Paas, 1998, p. 361). The probability which subjects possessing a different position on the latent trait (θ), in this case θ , can be a so-called information search effort, providing positive answers for each item.

Two requirements are needed to test whether probabilistic cumulative MSA exist. The first requirement is known as Monotone Homogeneity (MH), which tests whether a set of items measure a unidimensional latent trait (θ) (Mokken, 1971; Sijtsma & Molenaar, 2002; Van Schuur, 2011). In order to fulfil MH, the set of items must satisfy three assumptions: unidimensionality, non-decreasing monotonicity and stochastically independent responses (Mokken, 1971, 1997; Sijtsma & Molenaar, 2002). If the data satisfy these three MH assumptions, subjects can be ordered on a latent unidimensional (continuum) scale (Mokken, 1971; Van Schuur, 2011). Therefore, a total score (X_+) can be computed to order people on the latent scale (θ), thus an ordinal person scale is obtained.

The second criterion, which is a stricter requirement, is known as Double Monotonicity (DM). This criterion “requires identical ordering of negative and positive response probabilities on a set of items for all subjects” (Paas, 1998, p. 362). If the data satisfy the DM requirement, then the order of items is the same for the different groups of subjects (Mokken, 1971, 1997; Sijtsma & Molenaar, 2002).

3.2.1. Testing Monotone Homogeneity requirements

In order to test for unidimensionality, three different coefficients have to be computed. The first is the item-pair coefficients or Coefficient H_{jk} , which is a scalability coefficient for each pair of items used, j and k . The second is the item coefficient (H_j), which represents the scalability of one item in relation to the whole set of items on the scale. Items with high H_j values will contribute to creating item ordering. The third coefficient is the total-scale coefficient, computed using Loevinger's Homogeneity coefficient (H) (Loevinger, 1948; Mokken, 1971), which is the single score for the whole scale. This represents the degree to which the total score X_+ orders persons on the latent trait scale (θ), the higher H scores the more accurate person ordering on θ exists.

Unidimensionality of subsets of items requires a) positive item-pair coefficients (H_{jk}); b) values for $H_j > .30$, which means that the item belongs to a scale and it contributes to the person ordering (items with $H_j < .30$ will be discarded from the scale); and c) $H > .30$ (Mokken, 1971; Sijtsma & Molenaar, 2002). The method used to test monotonicity for MH is the rest-score method (Mokken, 1997; Van der Ark, 2007; 2012; Van Schuur, 2003).

3.2.2. Testing Double Monotonicity requirements

With MH items, a set of persons can be ranked in the same order on the continuum latent trait (θ). MH establishes a person order, however it is not sufficient to establish a uniform rank ordering of items, for instance, an ordering in which all subjects perceive the items with the same degree of popularity (Van Schuur, 2003, p. 150). To ensure the same ordering of items by subjects, Double Monotonicity (DM) is needed. This assumes non-intersection of the Item Response Functions (IRFs) for the different items.

Coefficient H^T is obtained for the set of items which DM holds. H^T expresses the degree of accuracy to which the order of the items

is the same for each respondent in the population. Ligtoet, van der Ark, te Marvelde, and Sijtsma (2010, p. 16) proposed the following classification: “ $H^T < .3$ means that the item ordering is too inaccurate to be useful; $.3 \leq H^T < .4$ means low accuracy; $.4 \leq H^T < .5$ means medium accuracy; and $H^T \geq .5$ means high accuracy”.

Finally, the reliability of the scale can be assessed. A common estimator for reliability is Cronbach's alpha, however it underestimates the test-score reliability (Sijtsma, 2009; Van der Ark, 2012), therefore “rho” or Molenaar and Sijtsma (MS) reliability statistic, which is an unbiased estimator of test-score reliability (Mokken, 1971; Molenaar & Sijtsma, 1984, 1988), is used. Table 2 shows a summary for testing MH and DM requirements.

3.2.3. Automated item selection procedure

In the case that all initial items do not have scalability, an automated item selection procedure is used, with the possibility of leaving some items unsuitable. In order to detect if a set of items are unidimensional, Mokken (1971) used a bottom-up hierarchical clustering algorithm, where it found the best smallest scale and then it adds the best items to the existing best smallest scale, until all scalable items have been incorporated into the scale (Van Schuur, 2003).

4. Results

Descriptive statistics (mean and standard deviation) for each source of information are shown in Table 3. The scale was measured by a 5-point Likert scale and for all items the minimum value was 1 and the maximum was 5. In general, the most frequently used information sources were clearly search engines, recommendations from friends and relatives, and the official website. While the least used were tourism fairs and travel agencies.

Mokken Scaling Analysis is carried out using the software package of statistical software R (Van der Ark, 2007; 2011, 2012). The unidimensionality on a cumulative scale is tested with Loevinger's H coefficient, which is satisfied if $H_{jk} > 0$, the score for each individual item (H_j) and the whole cumulative scale (H) is higher than .30 (Mokken, 1971). Thus, a set of items will be scalable and unidimensional if $H_{jk} > 0$, $H_j > .30$ and $H > .30$. A rule of thumb is that scales can be classified into weak scale ($.30 \leq H < .40$), medium scale ($.50 \leq H < .50$) and strong scale ($H \geq .50$) (Mokken, 1971, p. 185; Sijtsma & Molenaar, 2002, p. 60).

Results show that for all the items on information sources, a single unidimensional scale does not exist. Coefficient H for the whole scale is .317, which is acceptable. However, not all H_{jk} values are positive and H_j coefficients for the travel agency, tourism fairs, mass media and recommendations from friends and relatives are below the lower-bound .30. Table 3 shows the mean, standard deviation (sd), H_j coefficients and H_j 's standard errors (se).

Since these eight items are not scalable, the automated item selection procedure is carried out, obtaining two scales with

Table 2
Requirements for testing MH and DM.

Monotone homogeneity	
Unidimensionality	$H_{jk} > 0$ $H_j > .3$ $H > .30$
Monotonicity	$P(X_j \geq x \theta)$ Non-decreasing IRF function
Double Monotonicity	
Item Invariant Ordering	Non- Intersection of IRFs $H^T > .3$
Reliability	
MS Reliability	Generally Reliability $> .7$

Table 3
Descriptive statistics and H_j coefficients.

Information source	Mean	sd	H_j	se
Search engines	2.88	1.90	.54	.030
Official website	2.18	1.71	.43	.026
Tourism blogs	1.31	.89	.32	.035
Brochures	1.31	.88	.31	.033
Mass media	1.20	.69	.16	.033
Recommendation	2.40	1.74	.12	.035
Tourism fairs	1.01	.16	.12	.008
Travel agency	1.07	.47	.03	.085

*se = standard error; sd = standard deviation.

Table 4
Active and passive information search scales.

Information source	Scale	Active information search					Passive information search				
		Mean	H_j	Se	Zsig	Crit	Mean	H_j	se	Zsig	Crit
Search engines	1	2.88	.81	.032	0	0					
Official website	1	2.18	.79	.028	0	0					
Tourism blogs	1	1.31	.59	.049	0	0					
Brochures	1	1.31	.58	.049	0	0					
Recommendation	2						2.40	.67	.059	0	0
Mass media	2						1.20	.67	.059	0	12
Tourism fairs	NS*										
Travel agency	NS*										

*Not Scalable

scalable items. A first unidimensional scale (scale 1 in Table 4) is composed of four information sources (search engines, official website, tourism blogs and brochures) can be called “active information search”. The second unidimensional scale (scale 2 in Table 4) is made up of recommendation and mass media, which can be called “passive information search”. It is also significant that the items travel agency and tourism fairs are not scalable.

Table 4 shows that H coefficient for the whole scale of active information search is .72, all coefficients for each item are $H_j > .3$ and all H_{jk} coefficients are positive. For passive information search scale, H coefficient for the whole scale is .67, all $H_j > .3$ and all H_{jk} coefficients are positive. Therefore, since H coefficients for both scales are larger than .5, both scales can be considered as strong scales.

Monotonicity is tested using the rest score method (Mokken, 1971). Fig. 2 shows the probability of positive reactions (it means that all proportions are non-decreasing when the group score increases) for the different items on the active information search scale.

In addition to the graphical information, two statistical criteria are used for detecting non-decreasing IRF. One is the number of significant requirements not fulfilled (Zsig in Table 4). The other is a

general diagnostic statistic (Van der Ark, 2012, p. 10), called ‘crit value’ (crit in Table 4). High crit values (crit > 80) represent bad items (Van Schuur, 2011). Since all IRFs are non-decreasing, Zsig = 0 and crit < 80, the monotonicity requirement for active information search scale is fulfilled. The same requirements are fulfilled for the passive information search scale. MH requirements are fulfilled for both scales, which mean that subjects can be ordered on a latent unidimensional scale of both active and passive information search.

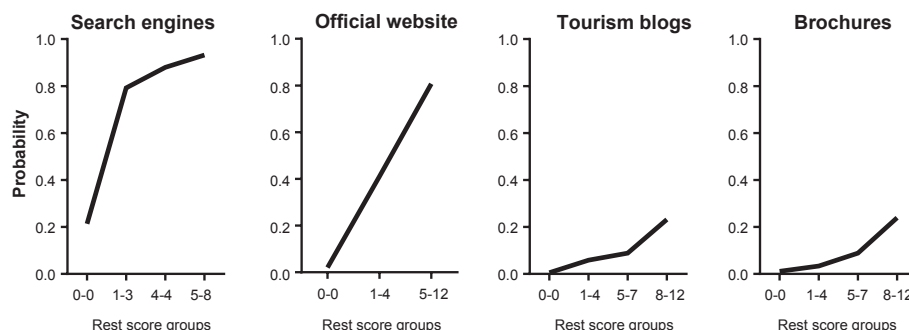
The next step is to test the Double Monotonicity (DM). This requires that the IRF's for the different items are not allowed to intersect. Fig. 3 shows an example of invariant item ordering for

search engines with brochures and official websites. In both cases items do not intersect. This pattern holds for the combination of each pair of items.

Statistical tests (Zsig and crit values) are also provided to check whether DM holds (Table 5). Results show that none of the items are problematic and H^T coefficient is .8, meaning high accuracy on the active information search scale. It can be concluded that the items on this scale can be ranked in the same order for all groups of subjects, from search engines ($H_j = .81$) to brochures ($H_j = .58$).

Non-intersection for passive information search is tested using the pmatrix method (Mokken, 1971, pp. 180–182; Sijtsma & Molenaar, 2002), as there are two items on the scale. This method is based on the fact that an item should be non-decreasing if non-intersection holds. Statistical tests are based on Zsig and crit values. Both values are equal to 0 for mass media and recommendation items. This means that DM also holds for the passive information search scale. Additionally, active information search has also been tested with pmatrix method and DM holds (all four items have a value equals to 0 on Zsig and crit statistics).

The MS reliability (Molenaar & Sijtsma, 1984, 1988) for active information search scale is .82. In order to compare it with a known method, Cronbach's alpha is obtained (.71). These results show a

**Fig. 2.** Monotonicity for items in active information search.

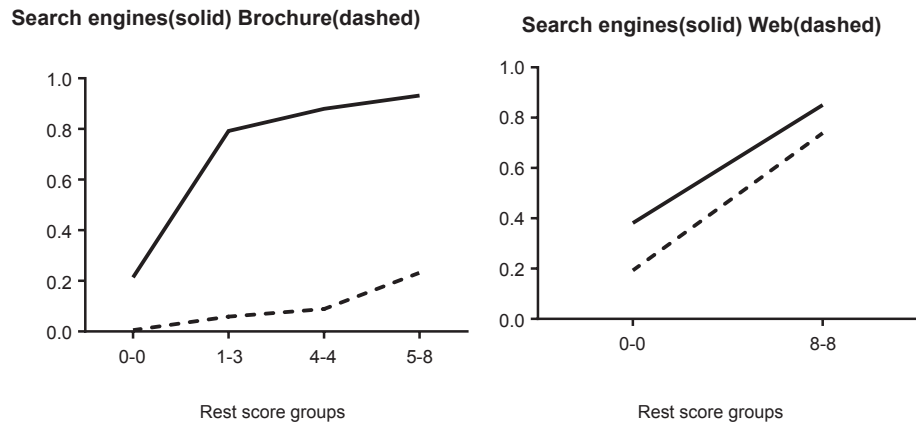


Fig. 3. Item invariant ordering.

Table 5
Double Monotonicity for active information search.

	H_i	Zsig	Crit
Search engines	.81	0	0
Official website	.79	0	0
Tourism blogs	.59	0	31
Brochures	.58	0	34
H^T	.80		

good reliability for the scale. In the case of passive information search, MS reliability is equal to .46 and Cronbach's alpha is .33. These low values are due to the fact that this scale is composed of only two items.

4.1. Scalability for socio-demographic and travel behavioural variables

An issue to address is whether the order of the items is constant when using different sample characteristics. In other words: "Is the scale patterns the same?" To find out, MSA will be used, taking into account socio-demographic (gender, age, place of residence, and education) and travel behavioural characteristics (accommodation, length of the stay and familiarity with the destination).

Similar to the whole sample, a unidimensional scale for information sources does not fulfil the requirements for monotonicity and scalability. Thus, results show a similar pattern between the categories of the different analysed characteristics. Table 6 summarizes the MH and DM for each socio-demographic characteristic. Reliability and the mean score for the unidimensional scale of active information search (Mean active in Table 6) and passive information search (Mean Passive in Table 6) are also reported.

Active and passive information search, are also found for the different socio-demographic characteristics. In general, active information search has the same structure and fits MH (from $H = .65$ to $H = .79$), meaning it is a strong scale. Items also fit DM (from $H^T = .59$ to $H^T = .86$), so there is high scale accuracy. Reliability is good, ranging from .77 to .85. When the mean scores obtained from the scale are compared, significant differences among socio-demographic characteristics are found. Males have significantly higher values than females, younger tourists also score higher than other ages; and international tourists have a higher score than domestic. In contrast, no differences were found between levels of education.

For all samples, MH and DM for passive information search are fulfilled. This is mainly composed of recommendations from friends

and relatives and mass media sources. There are, however, three exceptions: tourists over 55 years old; international tourists; and tourists with primary or secondary education, where the item travel agency also forms part of the scale. However, reliability for these scales is low due the fact that only two or three items make up the scale.

For travel behavioural characteristics (Table 7), a similar pattern in the whole sample is found. The same two scales, active and passive information search, were found. For active information search, MH (from $H = .56$ to $H = .85$) fits, meaning a strong scale. Items also fit DM (from $H^T = .72$ to $H^T = .92$), meaning high scale accuracy. In the same direction, reliability is good, ranging from .68 to .88. Scores for the cumulative scale are higher for tourists in hotels and for first time visitors. There are no differences regarding the length of stay at the destination.

Passive information search also fits MH and DM requirements. This is mainly composed of recommendations from friends and relatives and mass media sources. A different pattern is found for first-time visitors. The scale is created with mass media and travel agency, while recommendations from friends and relatives are not scalable. Reliability for passive information search is low, as it is for socio-demographic characteristics.

5. Conclusions

The purpose of this article was first to understand how information sources are used by tourists, not only to aggregate similar sources, but also to analyse if these information sources belong to a hierarchical unidimensional scale with ordered items; and second, to analyse the resulting scale for socio-demographic and travel behavioural variables, in order to study if scales can be differentiated according these variables.

Therefore, the main contribution of this paper is the demonstration that scalability of information sources by mean of MSA effectively exists. This method is commonly used to assess dimensionality and requirements for constructing scales in several fields (psychology, political science, marketing, medical research, etc.). However, the only application of this method in the tourism field is in the evaluation of cultural values, carried out by Mazanec et al. (2015). Consequently, this study uses MSA to evaluate tourism information search behaviour for the first time.

In this context, it is confirmed that scalability of information sources constitutes two different information search scales: active and passive information search. In particular, findings prove that active information search is composed of search engines, official websites, tourism blogs and brochures, which are ordered by

Table 6
MH and DM for socio-demographic characteristics.

	Gender				Age					
	Male (n = 243)		Female (n = 346)		Less than 44 (n = 248)		44 to 55 (n = 186)		More than 55 (n = 187)	
	Mean	H _j	Mean	H _j	Mean	H _j	Mean	H _j	Mean	H _j
Search engines	3.10	.85	2.71	.77	3.44	.74	2.76	.83	2.14	.86
Official website	2.34	.79	2.06	.78	2.51	.70	2.22	.82	1.65	.85
Tourism blogs	1.43	.55	1.21	.64	1.39	.55	1.35	.51	1.15	.80
Brochures	1.34	.59	1.28	.59	1.37	.50	1.33	.68	1.19	.53
Total H		.72		.72		.65		.74		.79
Total H ^T		.80		.79		.79		.78		.81
MS Reliability		.83		.82		.77		.83		.79
Mean Active	2.05*		1.81*		2.18*		1.91*		1.57*	
Recommendation	2.45	.61	2.36	.72	2.71	.61	2.04	.53	2.34	.76
Mass media	1.21	.61	1.19	.72	1.19	.61	1.07	.53	1.37	.69
Travel agency		–		–		–		–	1.08	.66
Total H		.61		.72		.61		.53		.71
Total H ^T		–		–		–		–		.96
MS Reliability		.42		.49		.43		.30		.60
Mean Passive	1.83		1.78		1.95*		1.55*		1.59	

	Place of residence				Education			
	National (n = 222)		International (n = 368)		Primary & Secondary (n = 251)		University (n = 338)	
	Mean	H _j	Mean	H _j	Mean	H _j	Mean	H _j
Search engines	2.07	.71	3.37	.84	2.95	.87	2.83	.77
Official website	1.69	.70	2.48	.81	2.27	.82	2.14	.77
Tourism blogs	1.18	.51	1.39	.60	1.34	.62	1.28	.59
Brochures	1.25	.55	1.34	.60	1.36	.67	1.27	.52
Total H		.64		.74		.77		.70
Total H ^T		.59		.86		.83		.78
MS Reliability		.81		.81		.85		.80
Mean Active	1.55*		2.14*		1.98		1.88	
Recommendation	2.34	.74	2.43	.58	2.39	.60	2.41	.72
Mass med	1.17	.74	1.21	.53	1.24	.55	1.17	.72
Travel agency		–	1.07	.32	1.06	.35		–
Total H		.74		.50		.53		.72
Total H ^T		–		.90		.90		–
MS Reliability		.43		.47		.45		.47
Mean Passive	1.75		1.82		1.56*		1.79*	

*p < 0.05.

popularity; and passive information search is composed of recommendations from friends and relatives and mass media, and are also ordered by popularity. In both cases, this indicates that, in general, tourists who use the less popular information sources also have used information sources, which are more popular.

This paper also contributes to academic literature by means of a number of findings related to the better understanding of information sources. It takes into consideration the effects of socio-demographic and travel behavioural variables, and particularly considers the two information search scales found.

First, regarding socio-demographic variables; gender, age, nationality and education have been considered. Findings prove that gender is a discriminating variable in terms of patterns of active information search. Males consult a higher number of sources than females, and it is demonstrated that differences in the preference of information sources exist, as proved by Luo et al. (2004). However, in the case of passive information search, no differences are identified. Such results diverge from Gitelson and Crompton (1983) who found that females are more likely to receive information from radio or television than males.

This study also concludes that age is a determining factor for information sources used, which is in line with previous research (Gronflaten, 2009; Luo et al., 2004). For instance, Fodness and Murray (1997, p. 513) found that “once children leave the home, the time and number of sources used to plan travel rises”. However, other studies do not agree with these findings. For instance,

Gronflaten (2009) found that age was not associated with information sources used, and Bruwer and Thach (2013) found that only word-of-mouth and brochures are influenced by age. Therefore, more research is needed in order to determine to what extent age affects scalability of information sources.

Nationality has also been postulated as a discriminant variable for active information search. International tourists tend to use a higher degree of information sources than domestic tourists. In this context, previous research has also confirmed that nationality influences information search behaviour (Gursoy & Chen, 2000; Money & Crotts, 2003).

Some particularities have been found regarding education. In the case of active information search, there are no differences among levels of education. The passive information search scale demonstrates that a lower level of education reduces the use of recommendations from friends and relatives and increases the use of mass media and travel agencies. This is partially in line with previous research, for instance, Gitelson and Crompton (1983) found that education level has effects on the number and type of information sources used and some types of sources. For example, in this study a relationship between level of education and print media was established.

Second, relating to travel behavioural variables; accommodation, length of stay and familiarity are considered influential variables. In particular, findings reveal that tourists who use hotels tend to consult more information sources, and are more likely to seek

Table 7
MH and DM for travel behavioural characteristics.

	Accommodation				Length of stay				Familiarity with destination			
	Hotel (n = 182)		No hotel (n = 368)		1 to 7 days (n = 298)		More than 7 (n = 295)		1 st visit (n = 211)		Repeat (n = 383)	
	Mean	H _j	Mean	H _j	Mean	H _j	Mean	H _j	Mean	H _j	Mean	H _j
Search engines	3.66	.83	2.52	.84	3.13	.68	2.62	.92	4.23	.57	2.13	.83
Official website	2.52	.80	1.98	.78	2.17	.65	2.18	.89	2.84	.65	1.82	.80
Tourism blogs	1.29	.66	1.27	.57	1.32	.46	1.30	.72	1.52	.43	1.19	.65
Brochures	1.19	.64	1.31	.59	1.30	.37	1.32	.78	1.39	.55	1.26	.59
Total H		.76		.73		.57		.85		.56		.75
Total H ^T		.92		.75		.74		.88		.86		.72
MS Reliability		.78		.84		.75		.88		.68		.86
Mean Active	2.17*		1.78*		1.98		1.85		2.50*		1.6*	
Recommendation	2.63	.61	2.27	.71	2.71	.66	2.08	.68	—	—	1.98	.74
Mass media	1.26	.61	1.18	.71	1.23	.66	1.16	.68	1.30	.33	1.14	.74
Travel agency		—		—		—		—	1.09	.33		—
Total H		.61		.71		.66		.68		.33		.74
Total H ^T		—		—		—		—		—		—
MS Reliability		.44		.48		.42		.49		.44		.50
Mean Passive	1.95*		1.65*		1.98*		1.62*		1.20		1.56	

*p < 0.05.

information using Internet in general (Gronflaten, 2009; Luo et al., 2004), and search engines in particular.

Length of stay is not an influential factor on information sources preferences for active information search. However, for passive information search, it is proved that tourists having shorter stays tend to be more influenced by recommendations from friends and relatives and mass media than those who have longer stays. Divergences with previous research are found. On one hand, some studies have concluded that higher levels of information search imply longer stays (Dale Fodness & Murray, 1999; Gitelson & Crompton, 1983); but on the other hand, Woodside and Ronkainen (1980) stated that there are no substantial differences in information sources used according to the length of the stay of the trip. Hence, considering that there are discrepancies, more research is needed in order to determine the effects of information search on the length of stay.

Previous research has concluded that familiarity with the destination influences both types and number of information sources used (Gursoy, 2003; Li et al., 2009). This is consistent with findings in this study, where tourists who travel to the destination for the first time use more active information sources than tourists who have visited the destination previously.

Findings demonstrate some discrepancies from previous research, especially considering that Bieger and Laesser (2004) argue that information search can be explained better by trip variables than by socio-demographic characteristics of the tourist. However, this is not completely supported in this study; mainly because some items included in both travel and socio-demographic characteristics have effects on information search behaviour. Therefore, taking into account that this study has only considered one set of variables, future research using an information sources scale should consider additional variables not available in this study. For example socio-demographic (i.e. household income, personal values and household status) and travel behavioural variables (i.e. travel party, planning horizon, purpose of trip, and so on).

Results in this study prove that information sources are scalable and follow an order in terms of popularity. Active and passive information search scales have been detected. At the same time, the resulting scales take into consideration the effect of segmenting variables. Therefore, these results are relevant in terms of marketing strategy development, in order to focus on targeted markets

according to their informational needs. “Communication strategies must focus primarily on those sources used most frequently by the targeted consumers” (D. Fodness & Murray, 1997, p. 520), taking into account both socio-demographic and travel behavioural variables.

5.1. Limitations and future research

As pointed out in previous research (i.e. Llodrà-Riera et al. 2015; Luo et al., 2004; Money & Crotts, 2003), several information sources should be considered. Taking into account that this study is not exhaustive in terms of the number of information sources included in the model, a longer list of information sources can be considered. Thus, future studies can include a number of information sources, in order to evaluate how items' scalability performs. In addition, Ho, Lin, and Chen (2012) highlight the current relevance of online information sources and evidence the need to better understand the information search process, including both online and offline information sources. Even though this study comprised a set of online information sources which are in the same scalable group, a further study including a wider range and higher number of online information sources can provide knowledge regarding how tourists use them. For example Llodrà-Riera et al. (2015) identified a continuum of 19 different online sources, which can be used as a basis.

Globally, this study provides a better understanding of tourists' behaviour in terms of information search effort. First, by determining the hierarchy of information search scale; and second, through the identification of differences in the behaviour of travellers based on socio-demographic and travel behavioural variables. Considering the demonstrated success in the application of MSA in information sources, future research can also use this method to analyse scalability in other fields of tourist behaviour analysis, such as motivation, perceived tourist image, preferences or attitudes. In fact, this is in line with suggestions provided by Mazanec et al. (2015, p. 303), who mention that knowledge of cultural values scale can contribute to “a better understanding of tourist motivations, information search and choice which are essential to market segmentation and tourism marketing strategy”.

References

- Bieger, T., & Laesser, C. (2004). Information sources for travel decisions: toward a source process model. *Journal of Travel Research*, 42(May), 357–371.

- Bruwer, J., & Thach, L. (2013). Wine tourists' use of sources of information when visiting a Usa wine region. *Journal of Vacation Marketing*, 19(3), 221–237.
- Buhalis, D., & Law, R. (2008). Progress in information technology and tourism management: 20 years on and 10 years after the Internet-The state of eTourism research. *Tourism Management*, 29, 609–623.
- Calantone, R. J., Di Benedetto, C. A., Hakam, A., & Bojanic, D. C. (1989). Multiple multinational tourism positioning using correspondence analysis. *Journal of Travel Research*, 28(2), 25–32.
- Camprubí, R., Guia, J., & Comas, J. (2013). The new role of tourists in destination image formation. *Current Issues in Tourism*, 16(2), 203–209.
- Dey, B., & Sarma, M. K. (2010). Information source usage among motive-based segments of travelers to newly emerging tourist destinations. *Tourism Management*, 31(3), 341–344.
- Embretson, S. E., & Reise, S. P. (2000). *Item response theory for psychologists*. Mahwah, NJ: Lawrence Erlbaum Associates.
- Emons, W. H. M., Sijtsma, K., & Pedersen, S. S. (2012). Dimensionality of the hospital anxiety and depression scale (hads) in cardiac patients: comparison of Mokken scale analysis and factor analysis. *Assessment*, 19, 337–353.
- Engel, J. F., Blackwell, R. D., & Miniard, P. W. (1995). *Consumer behaviour*. Fort Worth, TX: The Dryden Press.
- Fodness, D., & Murray, B. (1997). Tourist information search. *Annals of Tourism Research*, 24(3), 503–523.
- Fodness, D., & Murray, B. (1999). A model of tourist information search behavior. *Journal of Travel Research*, 37, 220–230.
- Gartner, W. C. (1994). Image formation process. *Journal of Travel and Tourism Marketing*, 2(3), 191–212.
- Gitelson, R. J., & Crompton, J. L. (1983). The planning horizons and sources of information used by pleasure vacationers. *Journal of Travel Research*, 21(3), 2–7.
- Gronflaten, O. (2009). Predicting travelers' choice of information sources and information channels. *Journal of Travel Research*, 48(2), 230–244.
- Gunn, C. A. (1972). *Vacationscape. Designing tourist regions*. Washington DC: Taylor and Francis/University of Texas.
- Gursoy, D. (2003). Prior product knowledge and its influence on the Traveler's information search behavior. *Journal of Hospitality & Leisure Marketing*, 10, 113–131.
- Gursoy, D., & Chen, J. S. (2000). Competitive analysis of cross cultural information search behavior. *Tourism Management*, 21(6), 583–590.
- Gursoy, D., & McCleary, K. W. (2004). An integrative model of tourists' information search behavior. *Annals of Tourism Research*, 31(2), 353–373.
- Guttman, L. (1950). The basis for scalogram analysis. In S. A. Stouffer, L. Guttman, E. A. Suchman, P. F. Lazarsfeld, S. A. Star, & J. A. Clausen (Eds.), *Measurement and prediction* (pp. 60–90). Princeton, NJ: Princeton University Press.
- Ho, C. I., Lin, M. H., & Chen, H. M. (2012). Web users' behavioural patterns of tourism information search: from online to offline. *Tourism Management*, 33(6), 1468–1482.
- Kerstetter, D., & Cho, M. H. (2004). Prior knowledge, credibility and information search. *Annals of Tourism Research*, 31(4), 961–985.
- Kim, H., & Richardson, S. L. (2003). Motion picture impacts on destination images. *Annals of Tourism Research*, 30(1), 216–237.
- Ligtvoet, R., van der Ark, L. A., te Marvelde, J. M., & Sijtsma, K. (2010). Investigating an invariant item ordering for polytomously scored items. *Educational and Psychological Measurement*, 70, 578–595.
- Li, X., Pan, B., Zhang, L., & Smith, W. W. (2009). The effect of online information search on image development: insights from a mixed-methods study. *Journal of Travel Research*, 48(1), 45–57.
- Llodrà-Riera, I., Martínez-Ruiz, M. P., Jiménez-Zarco, A. I., & Izquierdo-Yusta, A. (2015). A multidimensional analysis of the information sources construct and its relevance for destination image formation. *Tourism Management*, 48, 319–328.
- Loevinger, J. (1948). The technic of homogeneous tests compared with some aspects of scale analysis and factor analysis. *Psychological Bulletin*, 45, 507–529.
- Luo, M., Feng, R., & Cai, L. A. (2004). Information Search Behavior and Tourist Characteristics: the Internet vis-à-vis Other Information Sources. *Journal of Travel & Tourism Marketing*, 17(December 2013), 269–293.
- Mack, R. W., Blose, J. E., & Pan, B. (2008). Believe it or not: credibility of blogs in tourism. *Journal of Vacation Marketing*, 14, 133–144.
- Mazanec, J. A., Crotts, J. C., Gursoy, D., & Lu, L. (2015). Homogeneity versus heterogeneity of cultural values: an item-response theoretical approach applying Hofstede's cultural dimensions in a single nation. *Tourism Management*, 48, 299–304.
- Meijer, R. R., & Baneke, J. J. (2004). Analyzing psychopathology items: a case for nonparametric item response theory modeling. *Psychological Methods*, 9(3), 354–368.
- Mokken, R. J. (1971). *A theory and procedure of scale analysis*. The Hague: Mouton.
- Mokken, R. J. (1997). Nonparametric models for dichotomous responses. In W. Van der Linden, & R. K. Hambleton (Eds.), *Handbook of modern item response theory* (pp. 351–367). New York: Springer.
- Molenaar, I. W., & Sijtsma, K. (1984). Internal consistency and reliability in Mokken's nonparametric item response model. *Tijdschrift Voor Onderwijsresearch*, 9(5), 257–268.
- Molenaar, I. W., & Sijtsma, K. (1988). Mokken's approach to reliability estimation extended to multicategory items. *Kwantitatieve Methoden*, 9(2), 115–126.
- Money, R. B., & Crotts, J. C. (2003). The effect of uncertainty avoidance on information search, planning, and purchases of international travel vacations. *Tourism Management*, 24, 191–202.
- Nolan, S. D. (1976). Tourists' use and evaluation of travel information sources: summary and conclusions. *Journal of Travel Research*, 14(3), 6–8.
- Paas, L. J. (1998). Mokken scaling characteristic sets and acquisition patterns of durable- and financial products. *Journal of Economic Psychology*, 19, 353–376.
- Paas, L. J., & Molenaar, I. W. (2005). Analysis of acquisition patterns: a theoretical and empirical evaluation of alternative methods. *International Journal of Research in Marketing*, 22, 87–100.
- Roorda, L. D., Roebroek, M. E., van Tilburg, T., Molenaar, I. W., Lankhorst, G. J., & Bouter, L. M. (2005). Measuring activity limitations in walking: development of a hierarchical scale for patients with lower-extremity disorders who live at home. *Archives of Physical Medicine and Rehabilitation*, 86(12), 2277–2283.
- Sijtsma, K. (2009). On the use, the misuse, and the very limited usefulness of Cronbach's alpha. *Psychometrika*, 74, 107–120.
- Sijtsma, K., & Molenaar, I. (2002). *Introduction to nonparametric item response theory*. Thousand Oaks, CA: Sage Publications.
- Sirakaya, E., & Woodside, A. G. (2005). Building and testing theories of decision making by travellers. *Tourism Management*, 26, 815–832.
- Smits, I. A. M., Timmerman, M. E., & Meijer, R. R. (2012). Exploratory Mokken scale analysis as a dimensionality assessment tool: why scalability does not imply unidimensionality. *Applied Psychological Measurement*, 36, 516–539.
- Van Schuur, W. H. (2003). Mokken scale analysis: between the gutman scale and parametric item response theory. *Political Analysis*, 11, 139–163.
- Van Schuur, W. H. (2011). *Ordinal item response theory. Mokken Scale Analysis*. Thousand Oaks: Sage Publications.
- Van der Ark, L. A. (2007). Mokken scale analysis in R. *Journal of Statistical Software*, 20(11). Retrieved from <http://www.jstatsoft.org/v20/i11>.
- Van der Ark, L. A. (2011). *Getting started with Mokken scale analysis in R*.
- Van der Ark, L. A. (2012). New developments in mokken scale analysis in R. *Journal of Statistical Software*, 48(5). Retrieved from <http://www.jstatsoft.org/v48/i05/paper>.
- Van der Linden, W., & Hambleton, R. K. (1997). *Handbook of modern item response theory*. New York: Springer.
- Vargas-Vargas, M., Jimenez, J. M., Meseguer-Santamaria, M. L., Montoro-Lorenzo, J.-M., & Fernandez-Aviles, G. (2010). Measurement of students attitude towards statistics: a Mokken scales analysis of its dimensions. *Journal of Business Education*, 34(3), 1–6.
- Waller, N. G., Tellegen, A., McDonald, R. P., & Lykken, D. T. (1996). Exploring nonlinear models in personality assessment: development and preliminary validation of a negative emotionality scale. *Journal of Personality*, 64(3), 545–576.
- Walmsley, D. J., & Young, M. (1998). Evaluative images and tourism: the use of personal constructs to describe the structure of destination images. *Journal of Travel Research*, 36(3), 65–69.
- Wilson, T. D. (1997). Information behaviour: an interdisciplinary perspective. *Information Processing & Management*, 33(4), 551–572.
- Wismeijer, A. A. J., Sijtsma, K., van Assen, M. A. L. M., & Vingerhoets, A. J. J. M. (2008). A comparative study of the dimensionality of the self-concealment scale using principal components analysis and Mokken scale analysis. *Journal of Personality Assessment*, 90(4), 323–334.
- Woodside, A. G., & Ronkainen, I. A. (1980). Vacation planning segments: self planning vs. users of motor club and travel agents. *Annals of Tourism Research*, 7(3), 385–393.



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